
Geometry Problem Set

UTSA Problem Solving Club

This problem set is organized into three parts of increasing difficulty. **Part I** collects fundamental results that serve as building blocks—prove them from scratch even if they look familiar. **Part II** develops problem-solving techniques—reflections, loci, transformations, coordinates, and projective methods—that appear throughout mathematics: in linear algebra, group theory, differential geometry, and physics. **Part III** presents competition-level challenges requiring creativity and synthesis of ideas from the first two parts. Work through each part in order, but feel free to jump ahead to any problem that catches your eye.

Part I: Basic Geometry Facts

The results below are cornerstones of Euclidean and circle geometry. Even if a fact is familiar, proving it from scratch—ideally by more than one method—builds the geometric intuition needed for the later parts.

1. The Star's Points

Consider a five-pointed star (pentagram) formed by extending the sides of a convex pentagon.

Question: What is the sum of the five angles at the *tips* of the star?

Note: The star does not need to be regular!

Hint: Consider the exterior-angle theorem for triangles.

2. Triangle Area by Coordinates

Find the area of the triangle with vertices $A(1, 1)$, $B(7, 3)$, and $C(3, 9)$.

Hint: If $\vec{u} = B - A$ and $\vec{v} = C - A$, then the area of the triangle equals $\frac{1}{2}|\det[\vec{u} \ \vec{v}]|$. This is the Shoelace Formula.

3. The Midline Theorem

Let M and N be the midpoints of sides AB and AC of triangle ABC , respectively.

Prove that $MN \parallel BC$ and $MN = \frac{1}{2} BC$.

Hint: Write $\vec{MN}$ in terms of the position vectors of A , B , C .

4. Varignon's Parallelogram

Let $ABCD$ be *any* quadrilateral (not necessarily convex, and $ABCD$ need not even be planar!). Let P, Q, R, S be the midpoints of sides AB, BC, CD, DA , respectively.

Prove that $PQRS$ is a parallelogram.

Hint: Express $\vec{PQ}$ and $\vec{SR}$ in terms of the position vectors of A, B, C, D and use the Midline Theorem (Problem 3).

5. Angles and Chords

Four points A, B, C, D lie on a circle in that order. Chords AC and BD intersect at a point P inside the circle.

- (a) If $\angle BAC = 25^\circ$ and $\angle ACD = 35^\circ$, find $\angle BPC$.
- (b) State and prove the general formula for the angle formed by two chords that intersect inside a circle.

Hint: Each inscribed angle subtends an arc. Which arcs are relevant to the angle at P ?

6. Power of a Point

From a point P outside a circle, a tangent line touches the circle at T , and a secant line through P intersects the circle at points A and B (with A between P and B).

- (a) **Prove** that $PT^2 = PA \cdot PB$.
- (b) If $PT = 8$ and $PA = 4$, find the length AB .

Hint for (a): Look for a pair of similar triangles involving the tangent and the secant.

7. Ceva's Theorem

In triangle ABC , let D, E, F be points on sides BC, CA, AB respectively.

- (a) **Prove** that the cevians AD, BE, CF are concurrent if and only if

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1.$$

- (b) Use Ceva's theorem to give a one-line proof that the *medians* of a triangle are concurrent.
 (c) Use Ceva's theorem to prove that the *angle bisectors* of a triangle are concurrent.

Hint for (a): Compare the ratios of areas of triangles that share a common vertex.

8. Menelaus' Theorem

In triangle ABC , let X, Y, Z be points on lines BC, CA, AB respectively.

Prove that X, Y, Z are collinear if and only if

$$\frac{\overline{BX}}{\overline{XC}} \cdot \frac{\overline{CY}}{\overline{YA}} \cdot \frac{\overline{AZ}}{\overline{ZB}} = -1$$

where the ratios are *signed* (i.e. directed along each side).

Hint: The proof strategy parallels that of Ceva's theorem. Use areas with appropriate signs, or drop perpendiculars from the vertices to the line XYZ . Compare with Problem 7.

Part II: Training Exercises

Each problem below highlights a method that recurs across different areas of mathematics: reflection, parametric loci, coordinate proofs, vector algebra, geometric transformations, and a first taste of projective geometry. Focus on understanding *why* each technique works—the same ideas will reappear in linear algebra, group theory, differential equations, and physics.

9. The Shortest Path

Two points A and B lie on the same side of a line ℓ .

Question: Find the point P on ℓ that minimizes the total distance $AP + PB$, and prove that your construction gives the minimum.

Hint: Reflect one of the points across ℓ . What can you say about the straight line between a point and a reflection?

10. The Sliding Ladder

A rigid rod (“ladder”) of length L slides so that one end stays on the positive x -axis and the other on the positive y -axis. Let the angle the rod makes with the x -axis be θ .

- Show that the midpoint of the rod traces a quarter-circle of radius $L/2$.
- More generally, let P be the point that divides the rod in ratio $t : (1-t)$ from the x -end. Show that P traces an ellipse, and find its semi-axes.
- The mechanical device that produces this motion is the *trammel of Archimedes* (popularly called a “do-nothing machine”). Explain why every point on the rod—except the midpoint—traces an ellipse rather than a circle.

Hint: If the x -end is at $(L \cos \theta, 0)$ and the y -end at $(0, L \sin \theta)$, write the coordinates of the general point and eliminate θ .

11. The British Flag Theorem

Let $ABCD$ be a rectangle and let P be *any* point in the plane (not necessarily inside the rectangle).

Prove that

$$PA^2 + PC^2 = PB^2 + PD^2.$$

Hint: Place A at the origin with sides along the axes and compute both sides using coordinates.

12. The Euler Line

In any non-equilateral triangle, three distinguished points are:

- the **circumcenter** O (center of the circumscribed circle),
- the **centroid** G (intersection of the medians), and
- the **orthocenter** H (intersection of the altitudes).

- Prove that O, G, H are collinear. (The line through them is the *Euler line*.)
- Prove that G divides segment OH in the ratio $OG : GH = 1 : 2$.

Hint: Place the circumcenter at the origin, so $|\mathbf{a}|^2 = |\mathbf{b}|^2 = |\mathbf{c}|^2 = R^2$. Show that $H = \mathbf{a} + \mathbf{b} + \mathbf{c}$ and $G = (\mathbf{a} + \mathbf{b} + \mathbf{c})/3$.

13. The Apollonius Circle

Let A and B be two fixed points in the plane, and let $k > 0$ with $k \neq 1$.

- Show that the locus of all points P satisfying $PA/PB = k$ is a circle. (This is the *Apollonius circle* for A , B , and ratio k .)
- Find the center and radius of this circle in terms of A , B , and k .
- What happens when $k = 1$?

Hint: Set up coordinates with A and B on the x -axis. Square the condition $PA^2 = k^2 PB^2$ and expand.

14. Composition of Reflections

Let ℓ_1 and ℓ_2 be two lines in the plane that intersect at a point O , forming an angle θ .

- Prove that the composition “reflect in ℓ_1 , then reflect in ℓ_2 ” is a rotation about O through angle 2θ .
- What is the composition when $\ell_1 \parallel \ell_2$, separated by distance d ?
- The symmetries of an equilateral triangle are generated by reflections in any two of its three axes of symmetry. How many distinct symmetries are there? Describe them all and identify the group.

Hint for (a): If ℓ_1 makes angle α with the x -axis, the matrix of reflection in ℓ_1 is $\begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}$. Multiply the two reflection matrices.

15. Napoleon's Theorem

On each side of an *arbitrary* triangle ABC , construct an equilateral triangle on the exterior. Let P , Q , R be the centroids (equivalently, the circumcenters) of the three equilateral triangles.

Prove that PQR is itself an equilateral triangle.

Hint: Representing points in the plane as complex numbers, a 60° rotation about the origin is multiplication by $e^{i\pi/3}$. Alternatively, try 2×2 rotation matrices.

16. Homothety and the Nine-Point Circle

A *homothety* (dilation) with center C and ratio r sends each point P to the point P' on ray CP with $CP' = |r| \cdot CP$ (on the opposite side of C when $r < 0$).

- Show that the medial triangle of $\triangle ABC$ (formed by its three side-midpoints) is the image of $\triangle ABC$ under the homothety centered at the centroid G with ratio $-\frac{1}{2}$.
- Deduce that the circumcircle of the medial triangle (the *nine-point circle*) has center $N = \frac{1}{2}(O + H)$ and radius $R/2$, where O , H , R are the circumcenter, orthocenter, and circumradius of $\triangle ABC$.
- Show that the foot of each altitude of $\triangle ABC$ also lies on the nine-point circle.

Hint: For (b), apply the homothety from (a) to the circumcircle and use Problem 12 (Euler line). For (c), use coordinates with circumcenter at the origin.

17. Harmonic Division

Points C and D are *harmonic conjugates* with respect to A and B if C divides AB internally in some ratio $k : 1$ and D divides AB externally in the same ratio, i.e. $\frac{AC}{CB} = -\frac{AD}{DB}$ (signed ratios). Equivalently, the *cross-ratio* $(A, B; C, D) = -1$.

- In triangle PQR , the *internal* bisector of $\angle P$ meets QR at S , and the *external* bisector of $\angle P$ meets line QR at T . Show that S and T are harmonic conjugates with respect to Q and R .
- Let M be the midpoint of AB . Show that if $(A, B; C, D) = -1$, then $MA^2 = MC \cdot MD$ (using signed lengths).
- [Projective viewpoint.]** Four concurrent lines from a point O cut a transversal in a harmonic range. Show that they cut *every* transversal in a harmonic range. (This is the *projective invariance* of harmonic division.)

Hint for (a): Use the angle-bisector theorem ($QS/SR = PQ/PR$). For (b): place $A = -1$, $B = 1$ on a number line and show $CD = 1$.

Part III: Competition-Level Problems

These problems require creative combinations of the techniques from Parts I and II. Don't be discouraged if a problem takes a long time—some of the results below famously resisted quick solutions when they were first posed. For each problem, try at least two different approaches before looking at hints.

18. Ptolemy's Theorem

Let $ABCD$ be a cyclic quadrilateral (inscribed in a circle).

(a) **Prove** that

$$AC \cdot BD = AB \cdot CD + AD \cdot BC.$$

(b) By choosing four points on the unit circle at suitable angles, derive the identity $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$.

Hint for (a): Choose K on diagonal AC with $\angle ABK = \angle DBC$. Show that $\triangle ABK \sim \triangle DBC$ and $\triangle ABD \sim \triangle KBC$.

19. The Simson Line

Let P be a point on the circumcircle of triangle ABC . Drop perpendiculars from P to lines BC , CA , AB , with feet D , E , F respectively.

Prove that D , E , F are collinear. (The line DEF is the *Simson line* (or *pedal line*) of P with respect to $\triangle ABC$.)

Hint: Show that $\angle PDB + \angle PFB = 180^\circ$ (i.e. P , D , B , F are concyclic). Use the inscribed-angle theorem on the circumcircle to relate the angles at D , E , F .

20. The Isoperimetric Inequality for Triangles

Among all triangles with a given perimeter p , which has the largest area?

(a) Using Heron's formula and the AM–GM inequality, prove that the equilateral triangle maximizes the area.

(b) Conclude that for *any* triangle with perimeter p and area A :

$$A \leq \frac{p^2 \sqrt{3}}{36},$$

with equality if and only if the triangle is equilateral.

Hint: By Heron's formula, $A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = p/2$. Note that $(s-a) + (s-b) + (s-c) = s$.

21. The Envelope of a Sliding Rod

Return to the setting of Problem 10: a rod of length L slides with its endpoints on the positive x - and y -axes. As the rod moves, it sweeps out a family of line segments.

- (a) Write the equation of the line containing the rod when it makes angle θ with the x -axis.
- (b) The *envelope* of a one-parameter family of curves is the curve tangent to every member of the family. Find the envelope by solving

$$F(x, y, \theta) = 0 \quad \text{and} \quad \frac{\partial F}{\partial \theta}(x, y, \theta) = 0$$

simultaneously and eliminating θ .

- (c) Show that the envelope is the *astroid*

$$x^{2/3} + y^{2/3} = L^{2/3}.$$

Hint: The line through $(L \cos \theta, 0)$ and $(0, L \sin \theta)$ can be written as $x \sin \theta + y \cos \theta = \frac{L}{2} \sin 2\theta$. Differentiate with respect to θ and solve.

22. The Butterfly Theorem

Let M be the midpoint of a chord PQ of a circle. Through M , draw two other chords AB and CD (with A and D on the same side of PQ). Let $AD \cap PQ = X$ and $BC \cap PQ = Y$.

Prove that M is the midpoint of XY .

Hint: Set up coordinates with M at the origin and PQ along the x -axis. Use the equation of the circle to express X and Y , and show they are symmetric about the origin.

23. Monge's Theorem

Three circles $\omega_1, \omega_2, \omega_3$ with pairwise distinct radii are given in the plane. For each pair of circles, the two *external* common tangent lines meet at a point called the *external center of similitude*.

Prove that the three external centers of similitude are collinear.

Hint: The external center of similitude of two circles is the center of the unique homothety (with positive ratio) mapping one circle to the other. What happens when you compose two homotheties? Think about how Menelaus' theorem (Problem 8) might help.

24. Desargues' Theorem

Two triangles $A_1B_1C_1$ and $A_2B_2C_2$ are said to be *in perspective from a point* if lines A_1A_2 , B_1B_2 , C_1C_2 are concurrent (at a point called the *center of perspectivity*). They are *in perspective from a line* if the intersections of corresponding sides,

$$X = A_1B_1 \cap A_2B_2, \quad Y = B_1C_1 \cap B_2C_2, \quad Z = C_1A_1 \cap C_2A_2,$$

are collinear (on the *axis of perspectivity*).

Prove: Two triangles are in perspective from a point if and only if they are in perspective from a line.

Hint ("only if" direction): Apply Menelaus' theorem (Problem 8) to appropriate triangles formed by the configuration. For the converse, consider using coordinates or the dual statement.

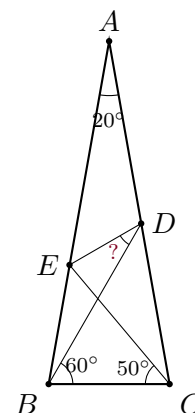
25. Langley's Adventitious Angles

Consider the isosceles triangle ABC with $\angle A = 20^\circ$ and $\angle B = \angle C = 80^\circ$. Points D and E are placed on sides AC and AB respectively, such that $\angle DBC = 60^\circ$ and $\angle ECB = 50^\circ$.

Question: Find the measure of $\angle BDE$.

Hint: This deceptively simple-looking problem stumped many professional mathematicians when first published. Direct angle-chasing alone won't work—try an auxiliary construction.

Source: E. M. Langley, Math. Gazette, 1922



26. No Equilateral Triangle on a Lattice

Prove that no equilateral triangle in the plane has all three vertices at lattice points (i.e. points with integer coordinates).

Hint: Suppose A, B, C are lattice points forming an equilateral triangle with side length s . Compute the area two ways: once using the Shoelace Formula (Problem 2), which gives a rational number, and once using the standard formula $\frac{s^2\sqrt{3}}{4}$. What does this imply about $\sqrt{3}$?

27. Euler's Formula for OI

Let $\triangle ABC$ have circumradius R , inradius r , circumcenter O , and incenter I .

Prove Euler's formula:

$$OI^2 = R^2 - 2Rr.$$

Hint: Let the incircle touch BC at X , and let line AI meet the circumcircle again at M . Show that M is the midpoint of arc BC (not containing A), and use Power of a Point (Problem 6) to show $MI = MB = MC$. Then compute OI^2 by the power of I with respect to the circumcircle.

28. The Erdős–Mordell Inequality

Let P be a point inside triangle ABC . Denote by R_A, R_B, R_C the distances from P to vertices A, B, C , and by d_a, d_b, d_c the distances from P to sides BC, CA, AB , respectively.

Prove that

$$R_A + R_B + R_C \geq 2(d_a + d_b + d_c),$$

with equality if and only if $\triangle ABC$ is equilateral and P is its center.

Hint: Show the stronger componentwise inequalities $R_A \geq d_b \frac{a}{c} + d_c \frac{a}{b}$ (and the two cyclic analogues), where a, b, c are the side lengths. Each follows from dropping a perpendicular from the foot of the altitude to the relevant sides. Then sum the three inequalities and apply AM–GM.

Source: P. Erdős, Amer. Math. Monthly, 1935

29. A Tangent and Two Chords

Chords AB and CD of a circle intersect at a point E inside the circle. Let M be an interior point of the segment EB . The tangent at E to the circle through D, E , and M intersects lines BC and AC at points F and G , respectively.

Find $\frac{EF}{EG}$ in terms of $t = \frac{AM}{AB}$.

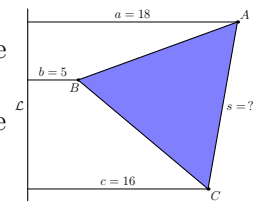
Hint: By the tangent-chord angle theorem applied to circle DEM , the tangent at E makes an angle with chord EM equal to the inscribed angle $\angle EDM$. Use the sine rule in triangles BEF and AEG , together with the inscribed angle theorem on the original circle.

Source: IMO 1990, Problem 1

30. An Implicit Lesson on Quadratic Forms

Let $\triangle ABC$ be an equilateral triangle and $\mathcal{L}$ a line on the plane.

Given the distances from the vertices A, B, C to $\mathcal{L}$, **find** the length s of the sides of $\triangle ABC$.



Source: Spelman College's Problem of the Month February 2017